

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460259.6822 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1715 +/- 0.0084
Transit depth (Rp/Rs)^2	2.94 +/- 0.29 [%]
Orbital Inclination (inc)	86.23 +/- 0.33
Airmass coefficient 1 (a1)	1.138 +/- 0.011
Airmass coefficient 2 (a2)	-0.08 +/- 0.027
Scatter in the residuals of the lightcurve fit is	0.99 %
Best Comparison Star	None
Optimal Aperture	3.44
Optimal Annulus	13.35
Transit Duration (day)	0.092 +/- 0.0075

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1715 ± 0.0084 vs 0.14195069566578392 (deviation 3.32σ)
Duration fit vs published	0.092 d vs 0.0875 d (deviation 0.57σ)
Mid-time O–C	8.9 min
Depth SNR	19.2
Residual scatter	0.99 %

Light curve

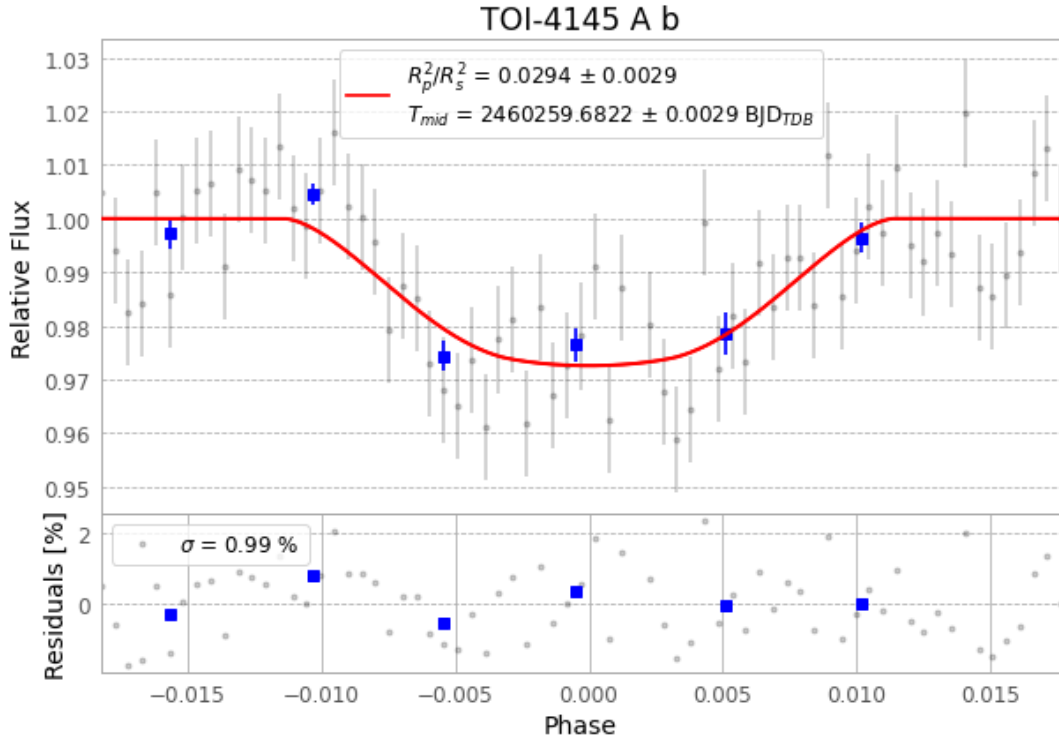


Figure 1 — FinalLightCurve_TOI-4145 A b_2023-11-10.png (SHA-256 2b352ed6541c924b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [329, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bd90c3f4d4bce02d...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

71 FITS frames (47 MB), TOI4145231111023016.FITS → TOI4145231111060015.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI4145-231111.log (cf774ba8eeb7...), FinalLightCurve_TOI-4145 A b_2023-11-10.png (2b352ed6541c...), FinalLightCurve_TOI-4145 A b_2023-11-10.csv (3f2a19d80a1a...)

Compute resources

Wall time 172.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-10 00:18Z.